

Time-Series Regression: A Practical Explainer

How to model relationships that unfold over time — and why it matters for housing markets

What Is a Time-Series?

A time-series is simply a sequence of observations collected at regular intervals over time — monthly housing prices, daily stock returns, annual GDP. Unlike a regular dataset where rows are independent, time-series data has a special property: each observation is related to the ones that came before it. Today's housing price is influenced by last month's housing price. This is called autocorrelation, and ignoring it leads to bad models.

Why Not Just Use Regular Linear Regression?

Standard linear regression assumes that your observations are independent of each other. In a time-series, that assumption is almost always violated. If you run a regular regression on monthly housing prices, your model will look good on paper but its standard errors will be wrong — meaning your p-values can't be trusted. Time-series regression methods account for this dependency explicitly.

The AR(1) Model

The simplest and most common time-series regression model is the AR(1) — Autoregressive model of order 1. It predicts today's value using yesterday's value as a predictor. The general form is:

$$Y_t = \alpha + \beta_1 X_t + \beta_2 Y_{t-1} + \text{error}_t$$

Where Y_t is your outcome today (e.g., housing price volatility this month), X_t is your predictor (e.g., policy uncertainty score), and Y_{t-1} is last month's outcome. The lagged term Y_{t-1} soaks up autocorrelation so the rest of your model is better specified.

How to Fit It in Python

Python's statsmodels library makes this straightforward. Here is a minimal example:

```
import statsmodels.api as sm
import pandas as pd
df['vol_lag1'] = df['price_volatility'].shift(1)
df_clean = df.dropna()
X = sm.add_constant(df_clean[['uncertainty_score', 'vol_lag1']])
y = df_clean['price_volatility']
model = sm.OLS(y, X).fit()
print(model.summary())
```

How to Interpret the Output

The model summary gives you several key numbers:

R-squared — How much of the variation in your outcome the model explains. A value of 0.32 means the model explains 32% of the variation in price volatility.

Adjusted R-squared — R-squared penalized for the number of predictors. With small samples, this drops sharply. An adjusted R^2 of 0.057 is common when $n=8$.

Coefficients (beta) — The estimated effect of each predictor. A positive β_1 means higher uncertainty is associated with higher volatility.

P-values — The probability of observing your coefficient by chance if the true effect is zero. Values below 0.05 are conventionally considered statistically significant.

A Note on Small Samples

Statistical tests lose power with fewer observations. With only 8 data points, you may find that your coefficients have the right sign and magnitude but are not statistically significant ($p > 0.05$). This does not mean your model is wrong — it means you need more data to be certain. In your report, focus on the direction and magnitude of the effect, not just the p-value, and be transparent about the limitation.

Reference: GeeksforGeeks — Time Series Regression.

<https://www.geeksforgeeks.org/data-science/time-series-regression/> | Prepared for DS 4002 Case Study, Spring 2026